Introduction to Data Visualization with Matplotlib

INTRODUCTION TO DATA VISUALIZATION WITH MATPLOTLIB



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datacamp

Data visualization

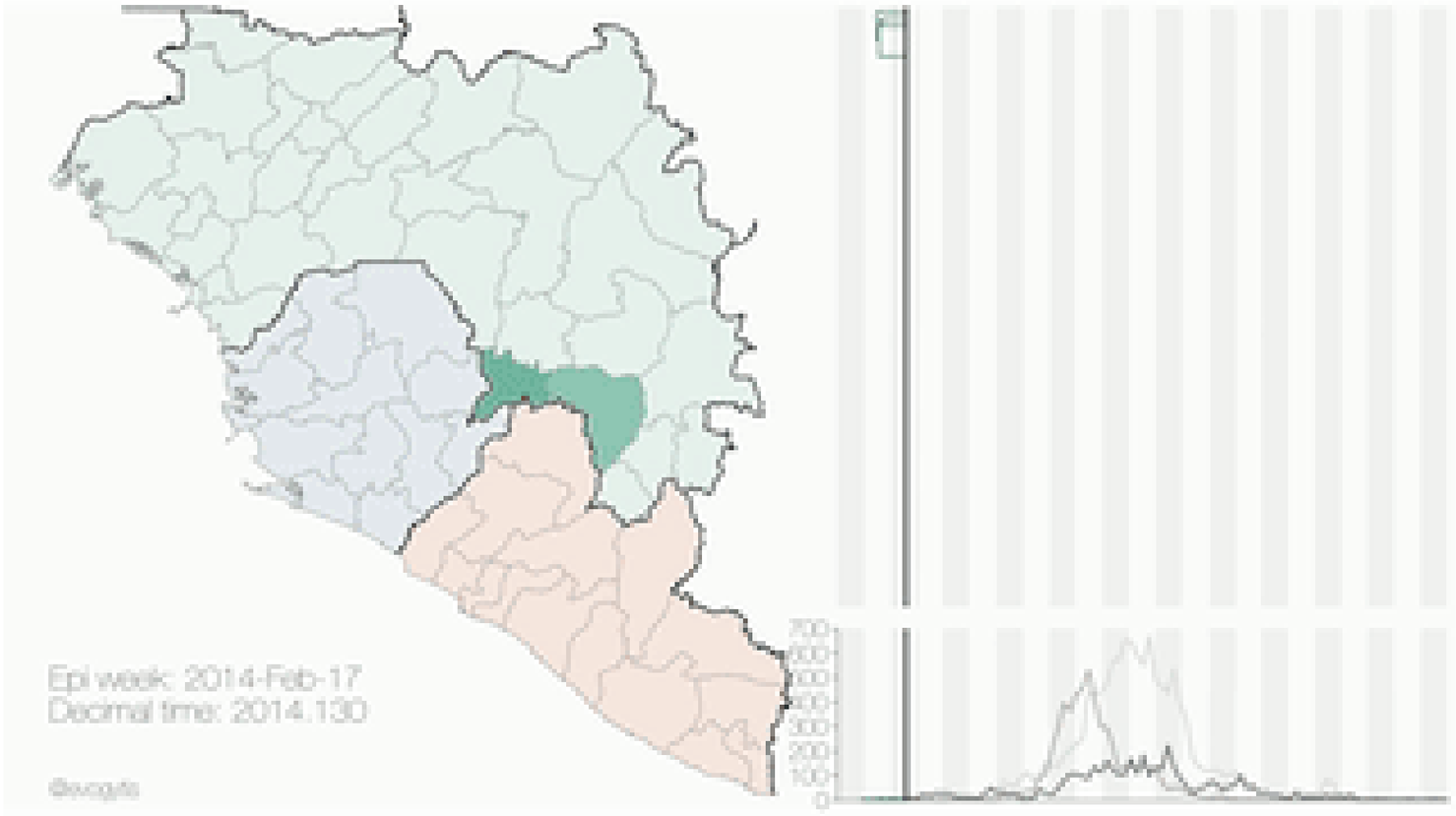
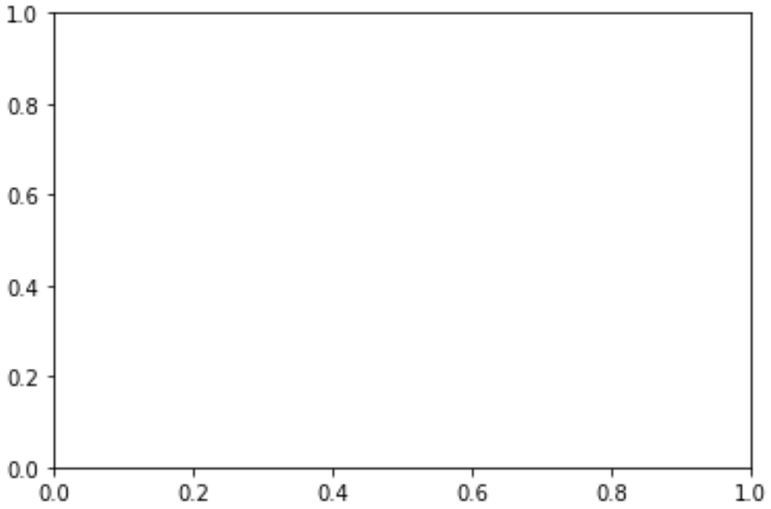


Image credit: **Gytis Dudas** and **Andrew Rambaut**

Introducing the pyplot interface

```
import matplotlib.pyplot as plt
fig, ax = plt.subplots()
plt.show()
```



Adding data to axes

```
seattle_weather["MONTH"]
```

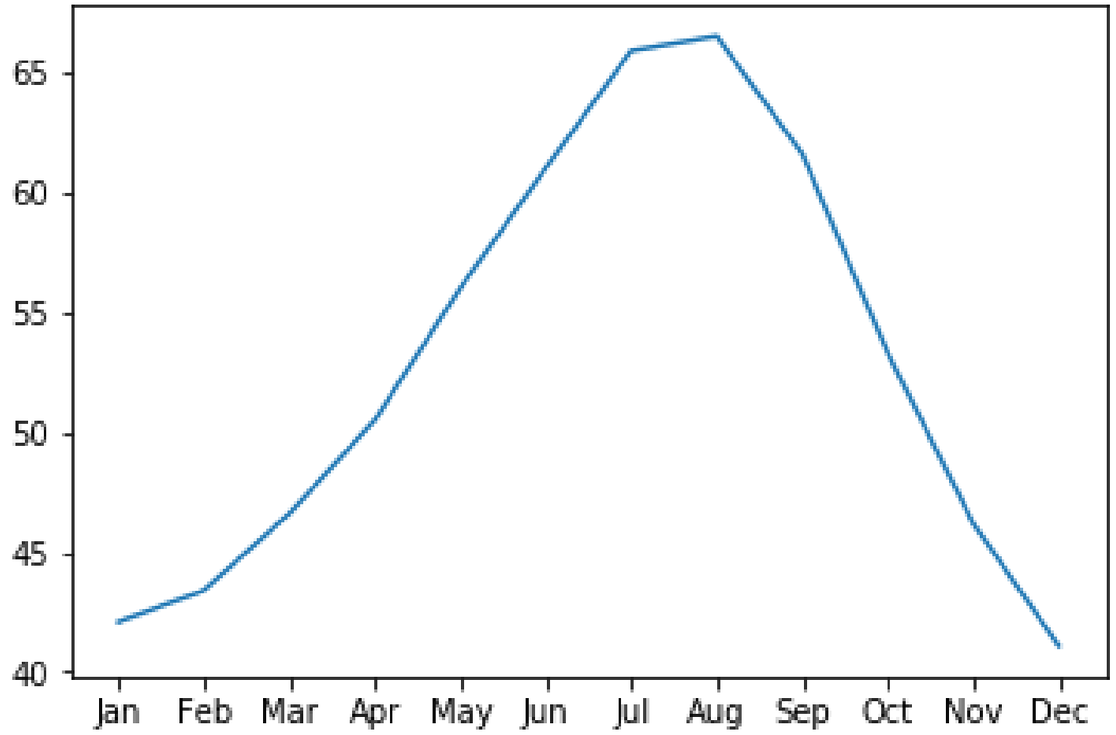
```
DATE
1     Jan
2     Feb
3     Mar
4     Apr
5     May
6     Jun
7     Jul
8     Aug
9     Sep
10    Oct
11    Nov
12    Dec
Name: MONTH, dtype: object
```

```
seattle_weather["MLY-TAVG-NORMAL"]
```

```
1     42.1
2     43.4
3     46.6
4     50.5
5     56.0
6     61.0
7     65.9
8     66.5
9     61.6
10    53.3
11    46.2
12    41.1
Name: MLY-TAVG-NORMAL, dtype: float64
```

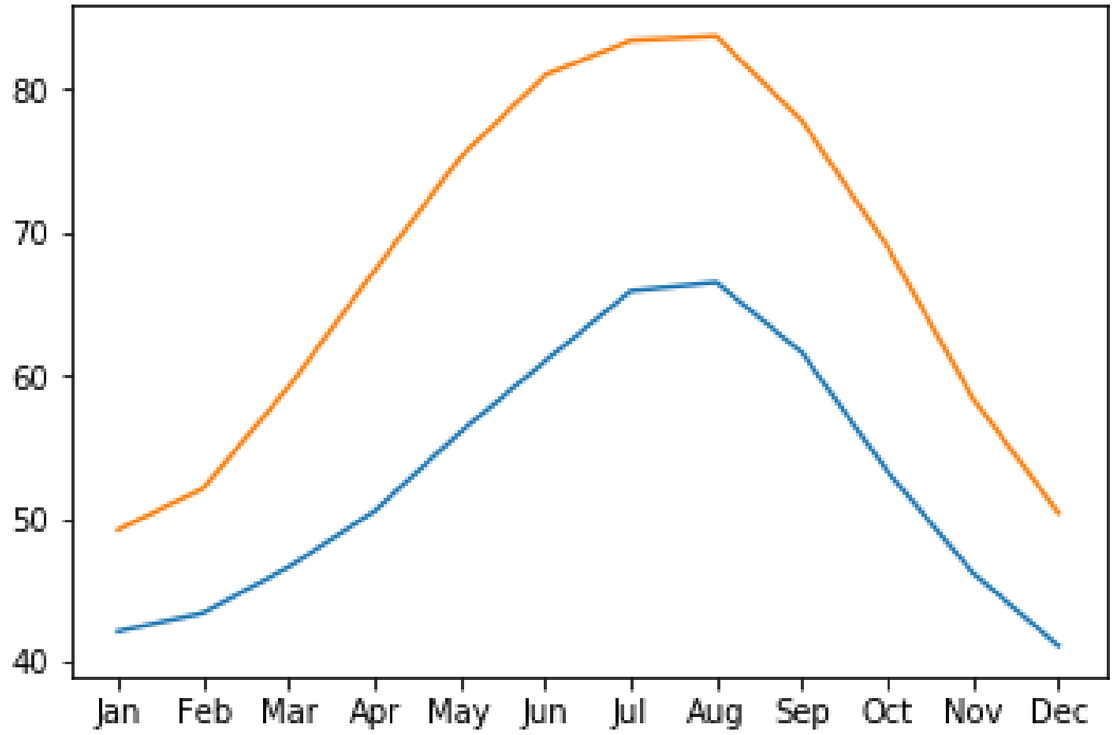
Adding data to axes

```
ax.plot(seattle_weather["MONTH"], seattle_weather["MLY-TAVG-NORMAL"])
plt.show()
```



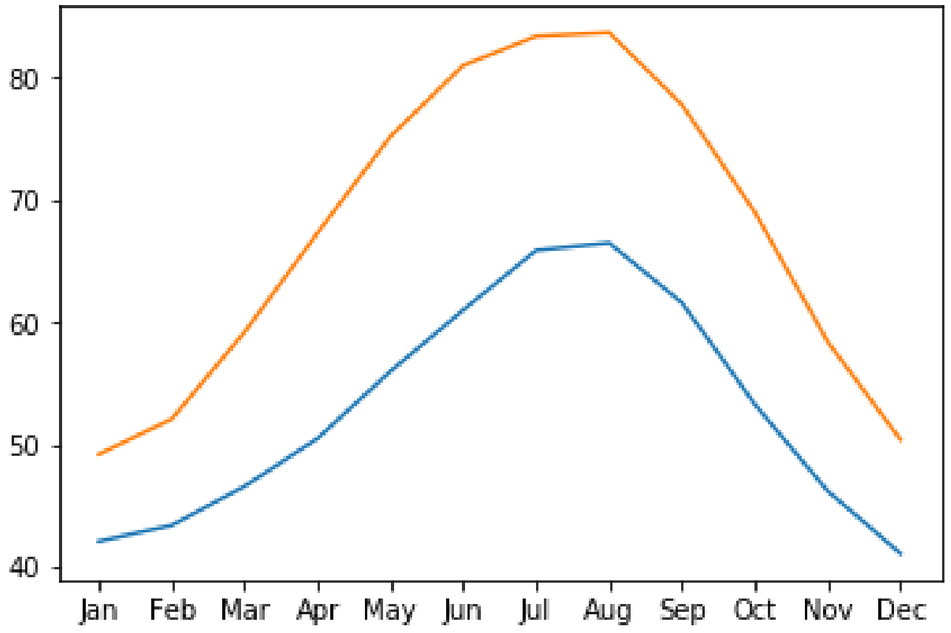
Adding more data

```
ax.plot(austin_weather["MONTH"], austin_weather["MLY-TAVG-NORMAL"])
plt.show()
```



Putting it all together

```
fig, ax = plt.subplots()
ax.plot(seattle_weather["MONTH"], seattle_weather["MLY-TAVG-NORMAL"])
ax.plot(austin_weather["MONTH"], austin_weather["MLY-TAVG-NORMAL"])
plt.show()
```



Practice making a figure!

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Customizing your plots

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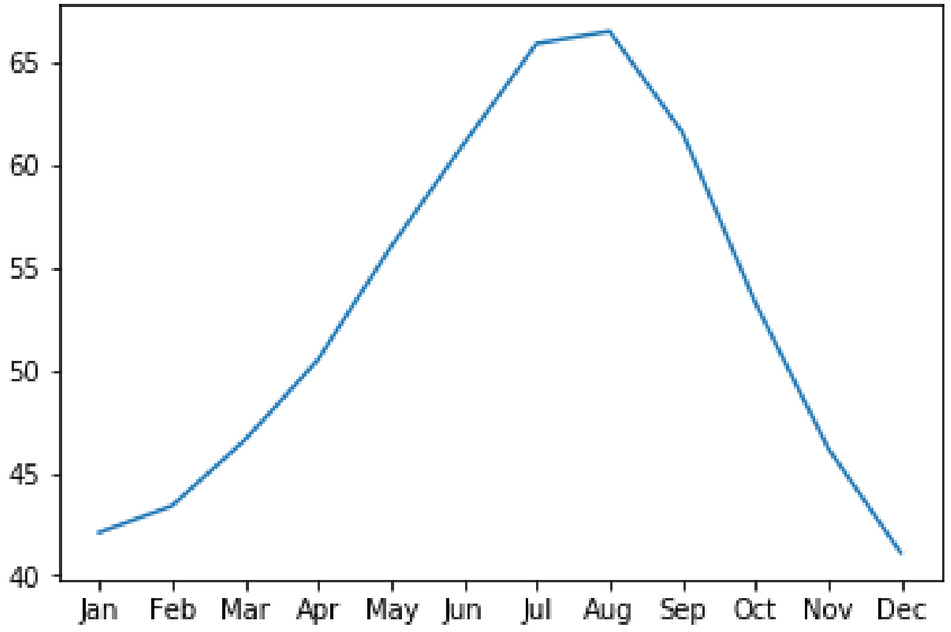


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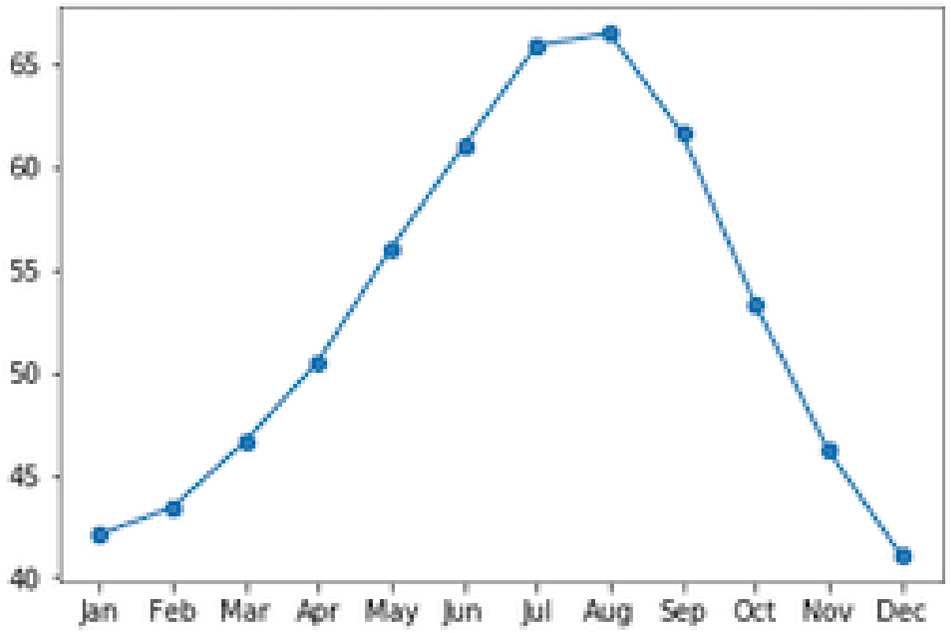
Customizing data appearance

```
ax.plot(seattle_weather["MONTH"],
        seattle_weather["MLY-PRCP-NORMAL"])
plt.show()
```



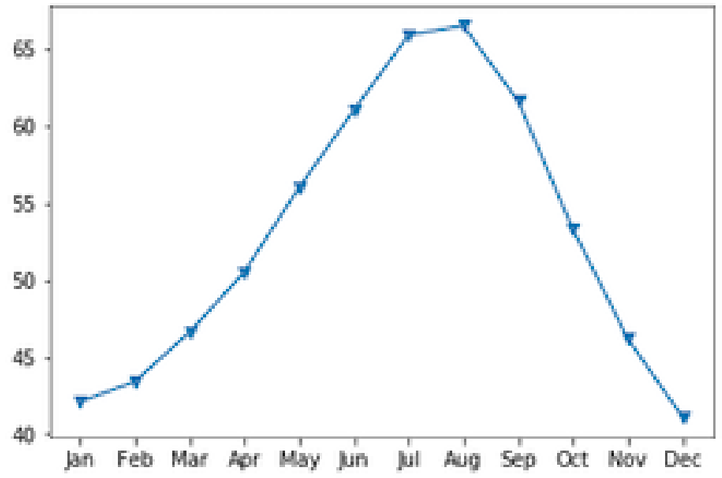
Adding markers

```
ax.plot(seattle_weather["MONTH"],
        seattle_weather["MLY-PRCP-NORMAL"],
        marker="o")
plt.show()
```



Choosing markers

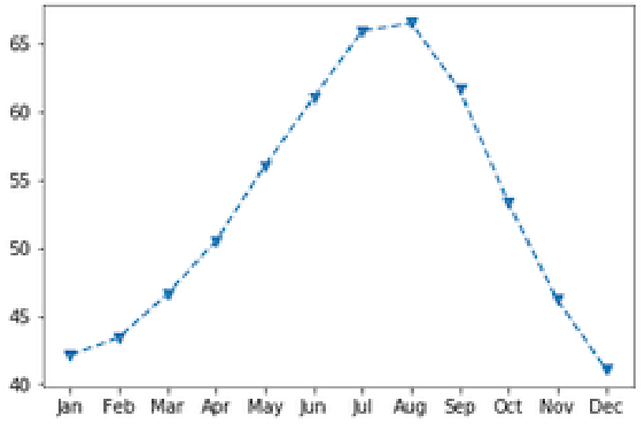
```
ax.plot(seattle_weather["MONTH"],
        seattle_weather["MLY-PRCP-NORMAL"],
        marker="v")
plt.show()
```



https://matplotlib.org/api/markers_api.html

Setting the linestyle

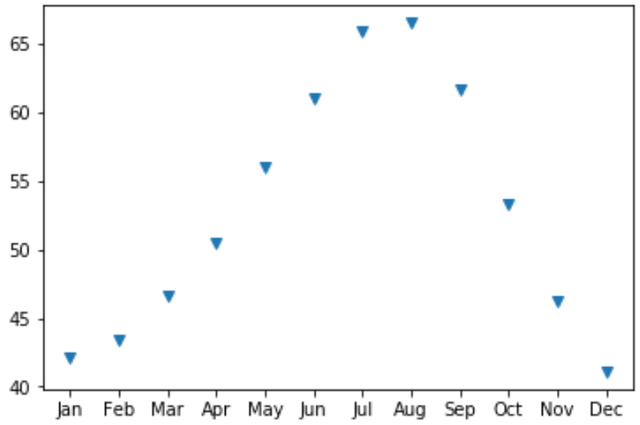
```
fig, ax = plt.subplots()
ax.plot(seattle_weather["MONTH"],
        seattle_weather["MLY-TAVG-NORMAL"],
        marker="v", linestyle="--")
plt.show()
```



https://matplotlib.org/gallery/lines_bars_and_markers/line_styles_reference.html

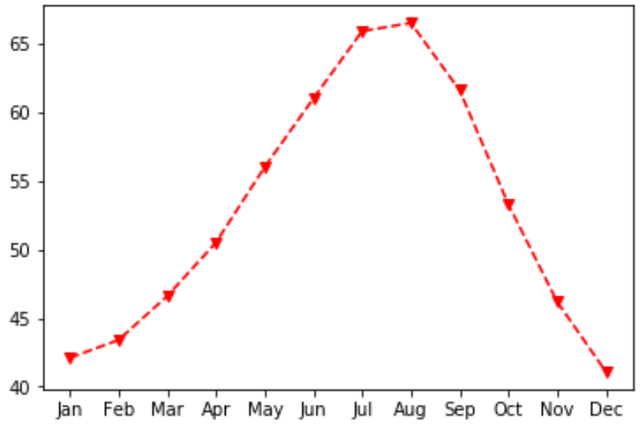
Eliminating lines with linestyle

```
fig, ax = plt.subplots()
ax.plot(seattle_weather["MONTH"],
        seattle_weather["MLY-TAVG-NORMAL"],
        marker="v", linestyle="None")
plt.show()
```



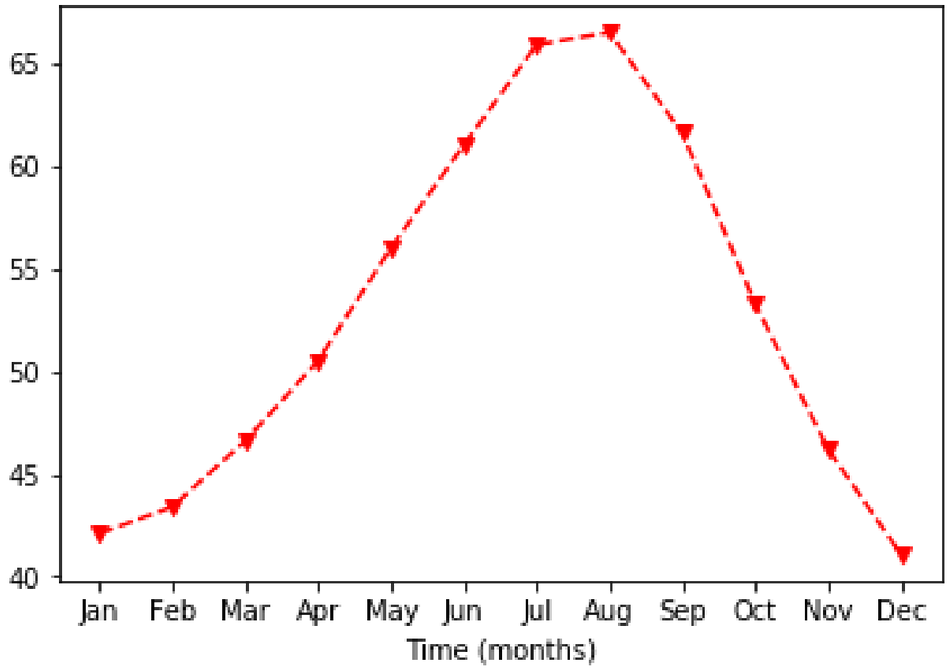
Choosing color

```
fig, ax = plt.subplots()
ax.plot(seattle_weather["MONTH"],
        seattle_weather["MLY-TAVG-NORMAL"],
        marker="v", linestyle="--", color="r")
plt.show()
```



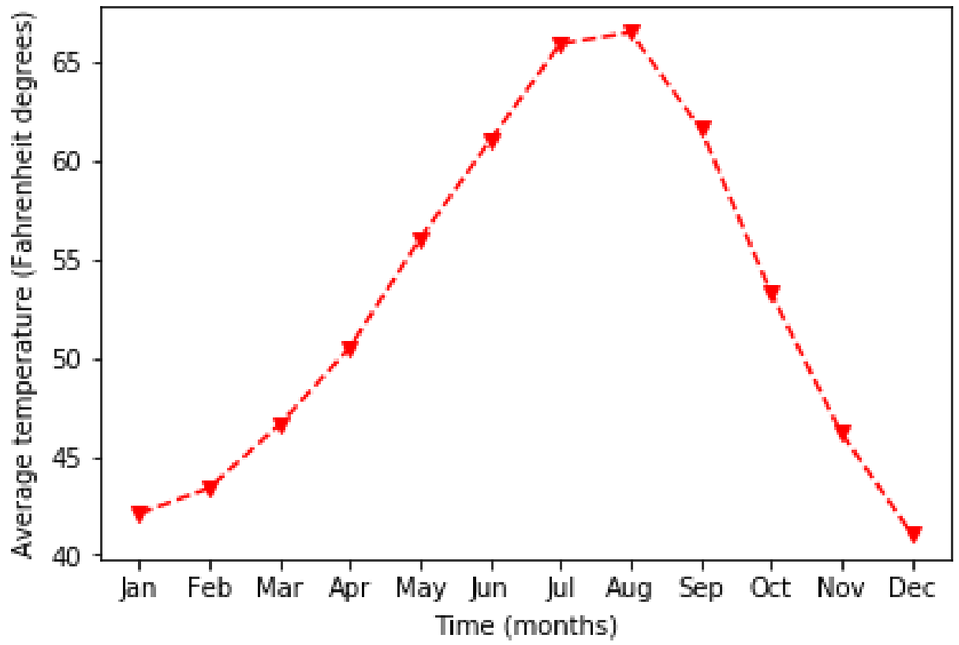
Customizing the axes labels

```
ax.set_xlabel("Time (months)")
plt.show()
```



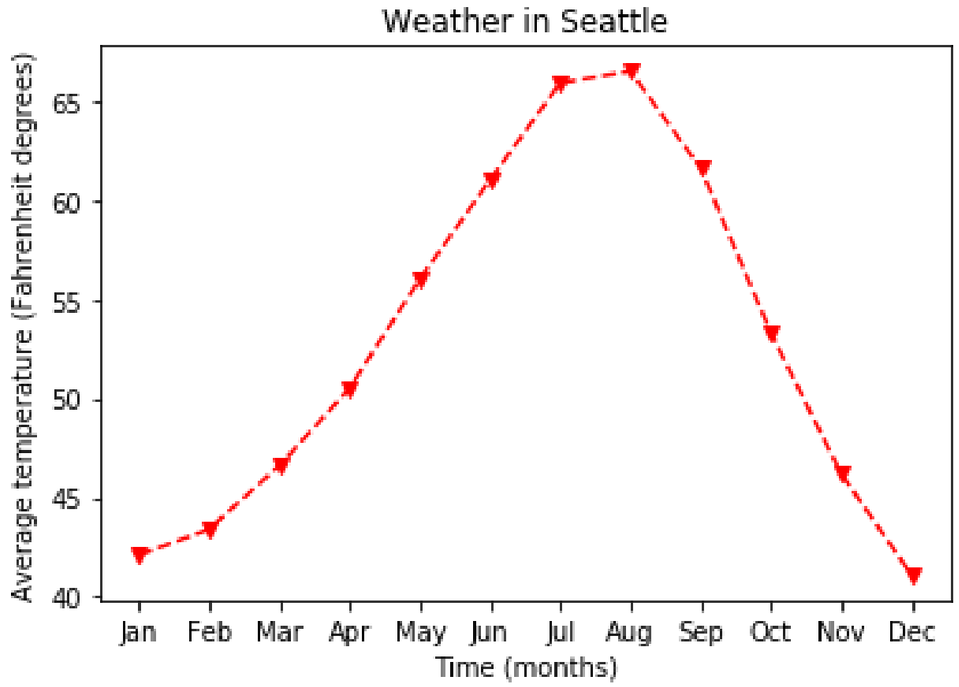
Setting the y axis label

```
ax.set_xlabel("Time (months)")
ax.set_ylabel("Average temperature (Fahrenheit degrees)")
plt.show()
```



Adding a title

```
ax.set_title("Weather in Seattle")
plt.show()
```



Practice customizing your plots!

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Small multiples

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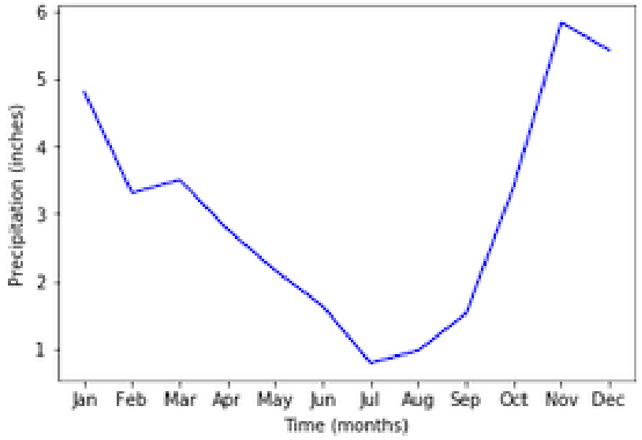


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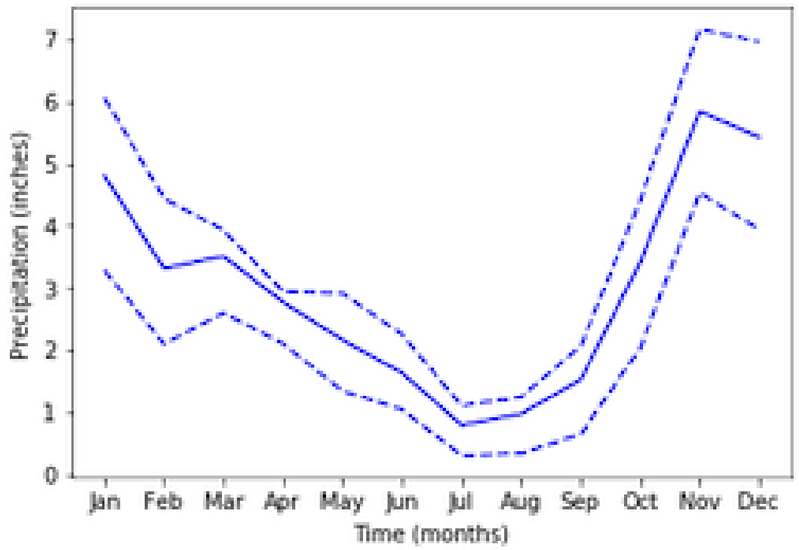
Adding data

```
ax.plot(seattle_weather["MONTH"],
        seattle_weather["MLY-PRCP-NORMAL"],
        color='b')
ax.set_xlabel("Time (months)")
ax.set_ylabel("Precipitation (inches)")
plt.show()
```



Adding more data

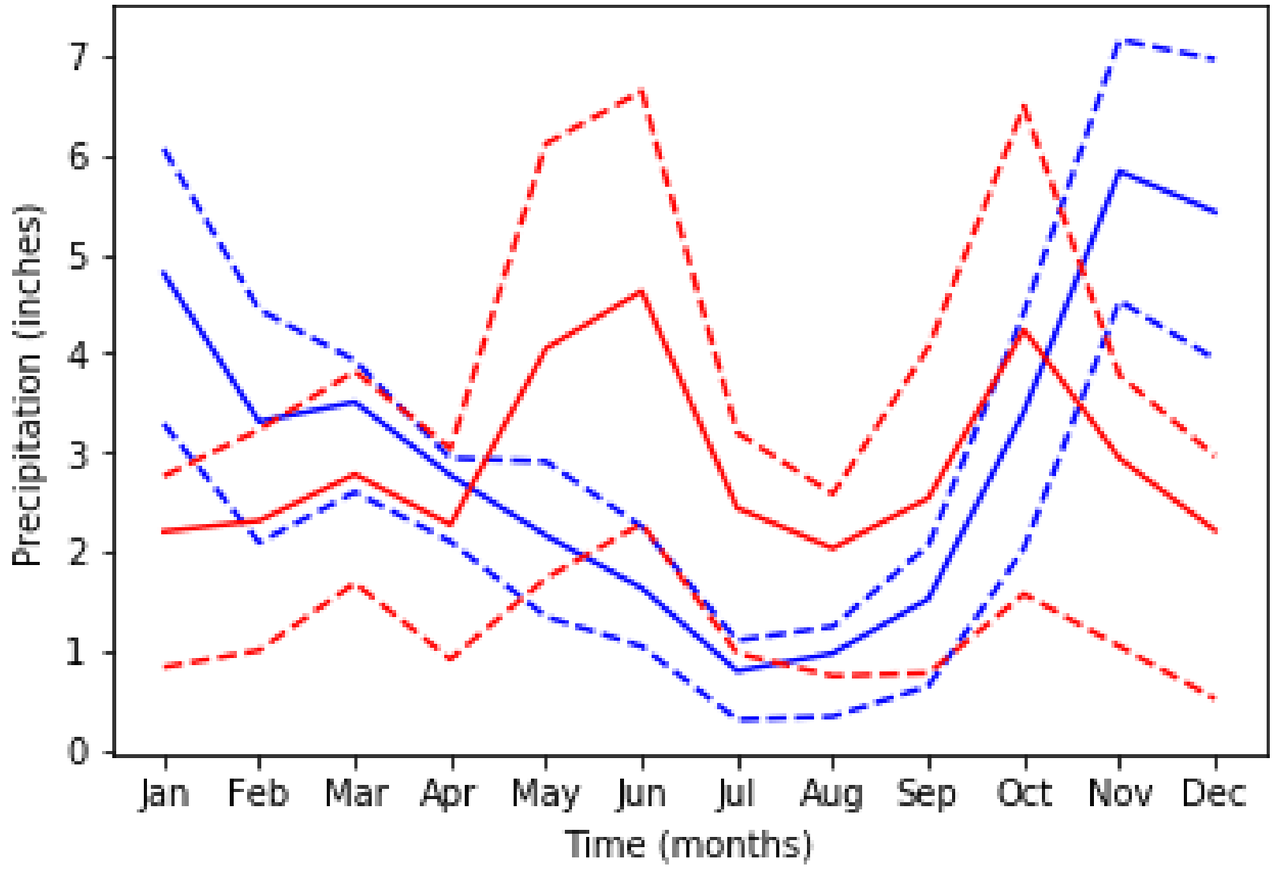
```
ax.plot(seattle_weather["MONTH"], seattle_weather["MLY-PRCP-25PCTL"],
        linestyle='--', color='b')
ax.plot(seattle_weather["MONTH"], seattle_weather["MLY-PRCP-75PCTL"],
        linestyle='--', color=color)
plt.show()
```



And more data

```
ax.plot(austin_weather["MONTH"], austin_weather["MLY-PRCP-NORMAL"],
        color='r')
ax.plot(austin_weather["MONTH"], austin_weather["MLY-PRCP-25PCTL"],
        linestyle='--', color='r')
ax.plot(austin_weather["MONTH"], austin_weather["MLY-PRCP-75PCTL"],
        linestyle='--', color='r')
plt.show()
```

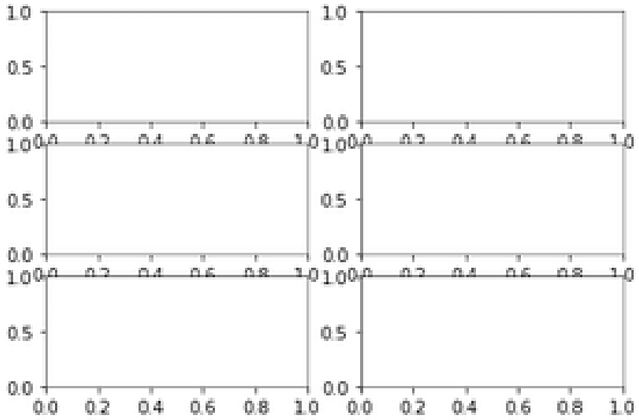
Too much data!



Small multiples with plt.subplots

```
fig, ax = plt.subplots()
```

```
fig, ax = plt.subplots(3, 2)
plt.show()
```

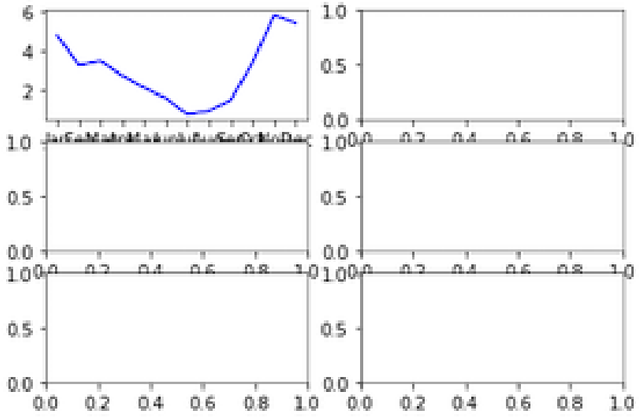


Adding data to subplots

```
ax.shape
```

```
(3, 2)
```

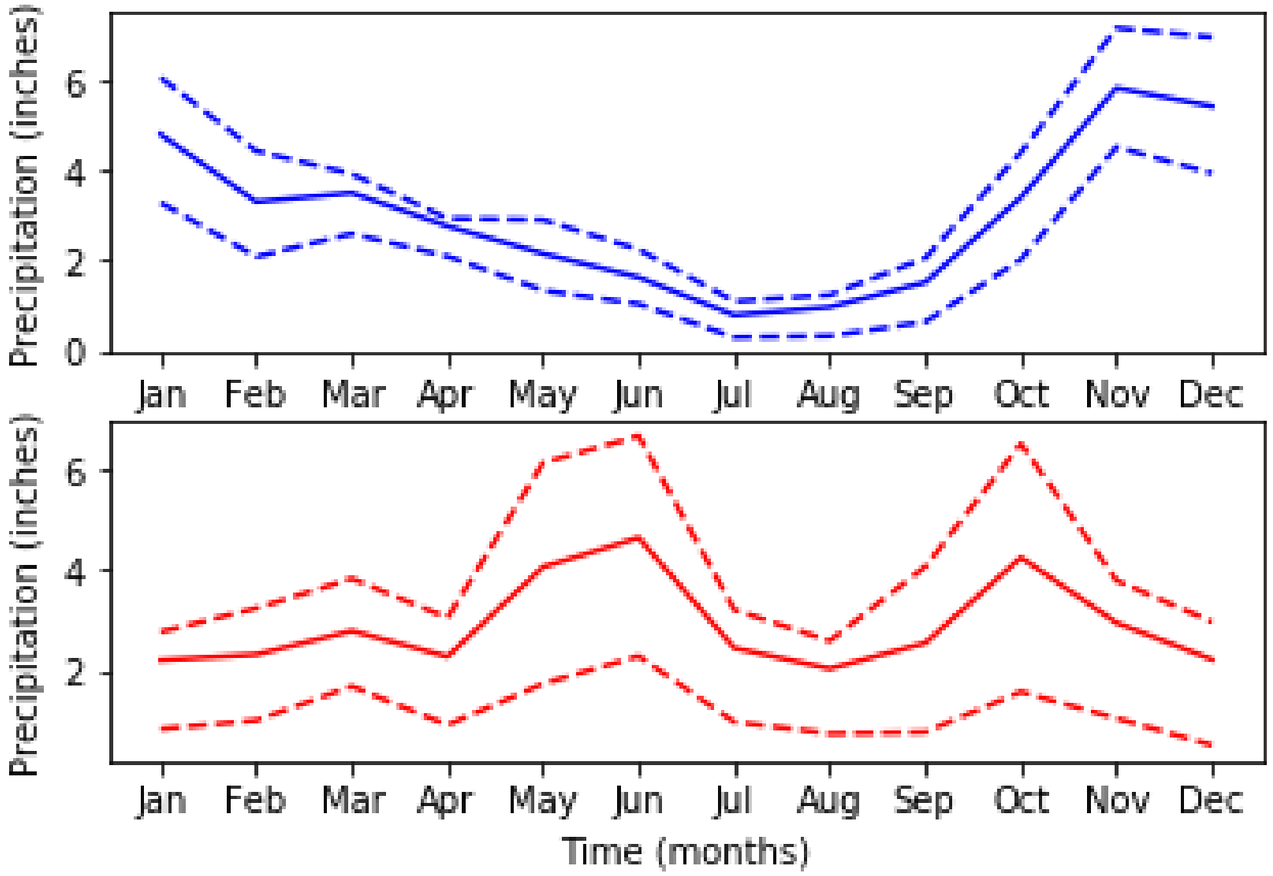
```
ax[0, 0].plot(seattle_weather["MONTH"],
              seattle_weather["MLY-PRCP-NORMAL"],
              color='b')
plt.show()
```



Subplots with data

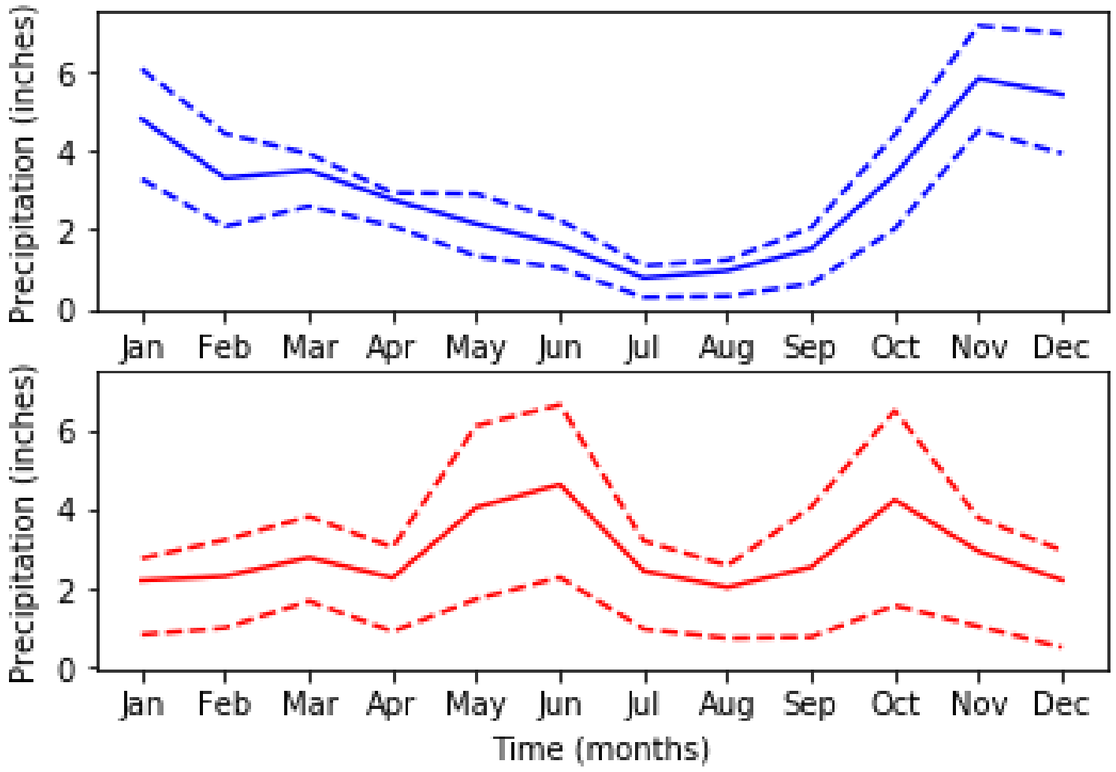
```
fig, ax = plt.subplots(2, 1)
ax[0].plot(seattle_weather["MONTH"], seattle_weather["MLY-PRCP-NORMAL"],
           color='b')
ax[0].plot(seattle_weather["MONTH"], seattle_weather["MLY-PRCP-25PCTL"],
           linestyle='--', color='b')
ax[0].plot(seattle_weather["MONTH"], seattle_weather["MLY-PRCP-75PCTL"],
           linestyle='--', color='b')
ax[1].plot(austin_weather["MONTH"], austin_weather["MLY-PRCP-NORMAL"],
           color='r')
ax[1].plot(austin_weather["MONTH"], austin_weather["MLY-PRCP-25PCTL"],
           linestyle='--', color='r')
ax[1].plot(austin_weather["MONTH"], austin_weather["MLY-PRCP-75PCTL"],
           linestyle='--', color='r')
ax[0].set_ylabel("Precipitation (inches)")
ax[1].set_ylabel("Precipitation (inches)")
ax[1].set_xlabel("Time (months)")
plt.show()
```

Subplots with data



Sharing the y-axis range

```
fig, ax = plt.subplots(2, 1, sharey=True)
```



Practice making subplots!

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